

MSC BUSINESS ANALYTICS • TEXT ANALYTICS GROUP PROJECT

Emotion Detection in Text

A Natural Language Processing & Machine Learning Approach Using BERT Embeddings

Group 2



Dataset: dair-ai/emotion (Hugging Face) • Deployment: Streamlit Community Cloud

<https://emotion-detection-ayr8nn66wjtuzydbkezftw.streamlit.app/>

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1. Background

Why detect emotion, not just sentiment?



Beyond positive / negative

Sentiment analysis collapses nuance; emotion labels (joy, anger, fear...) reveal why customers feel a certain way.



Scale of unstructured text

Organisations receive thousands of reviews, tickets and social posts daily — manual tagging cannot keep up.



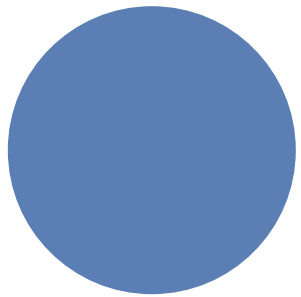
The assigned dataset

dair-ai/emotion: ~20,000 English tweets labelled with 6 basic emotions (Saravia et al., 2018).

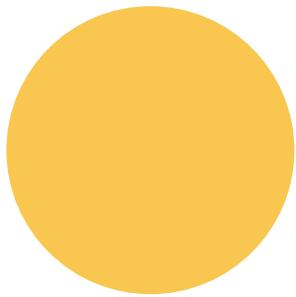
Objective: automatically classify short English text into sadness, joy, love, anger, fear, or surprise — then deploy the model as a live web app.

2. The Dataset — dair-ai/emotion

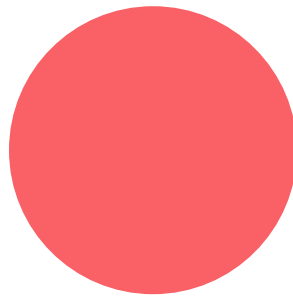
English-language tweets, each labelled with exactly one emotion. Pre-split into train / validation / test.



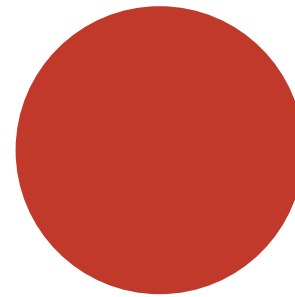
Sadness



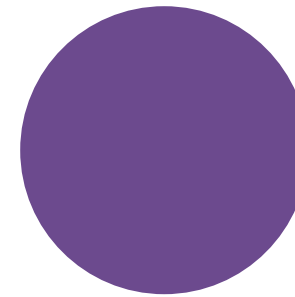
Joy



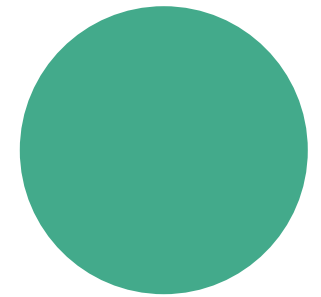
Love



Anger



Fear



Surprise

Six emotion classes — joy and sadness dominate; love and surprise are minority classes, motivating macro-averaged metrics over raw accuracy.

Splits

— Train ~16,000 • Validation ~2,000 • Test ~2,000.

3. Methodology — End-to-End Pipeline



Raw tweets → regex cleaning → BERT embeddings (768-dim) → Logistic Regression / SVM → evaluation → Streamlit app.

3.1 Preprocessing & Text Representation



Cleaning (Regex)

Remove URLs:

`http\S+`

Strip non-letters:

`[^a-zA-Z\s]`

Lower-case & trim whitespace



BERT Embeddings

bert-base-uncased

— 12 layers, 768-dim hidden state

Mean-pool last hidden state → one 768-dim vector per tweet

Max length 128 tokens, batch size 32

Captures context BoW / TF-IDF cannot (e.g. sarcasm, idioms)

3.2 Two classifiers, one embedding space



Logistic Regression

- Linear model, max_iter = 1000
- Fast to train, easy to interpret
- Strong baseline on dense embeddings



SVM (linear kernel)

- probability=True for confidence scores
- Strong on high-dimensional, separable data
- Typically slower to train

Both models are trained and evaluated on identical BERT embeddings — isolating the comparison to algorithm choice alone.

3.3 Evaluation Metrics



Accuracy

Overall proportion of correct predictions



Precision / Recall

Per-class correctness vs. completeness, macro-averaged



F1-score

Harmonic mean of precision & recall



Confusion Matrix

Which emotions get confused with which



ROC-AUC (macro, OvR)

Threshold-independent class separability

4. Results

Test-set performance from the group's actual training run (200 test examples).

Metric	Logistic Regression	SVM
Accuracy	0.500	0.530
Precision (macro)	0.375	0.428
Recall (macro)	0.349	0.378
F1-score (macro)	0.346	0.388
ROC-AUC (macro, OvR)	0.790	0.797

Best model: SVM

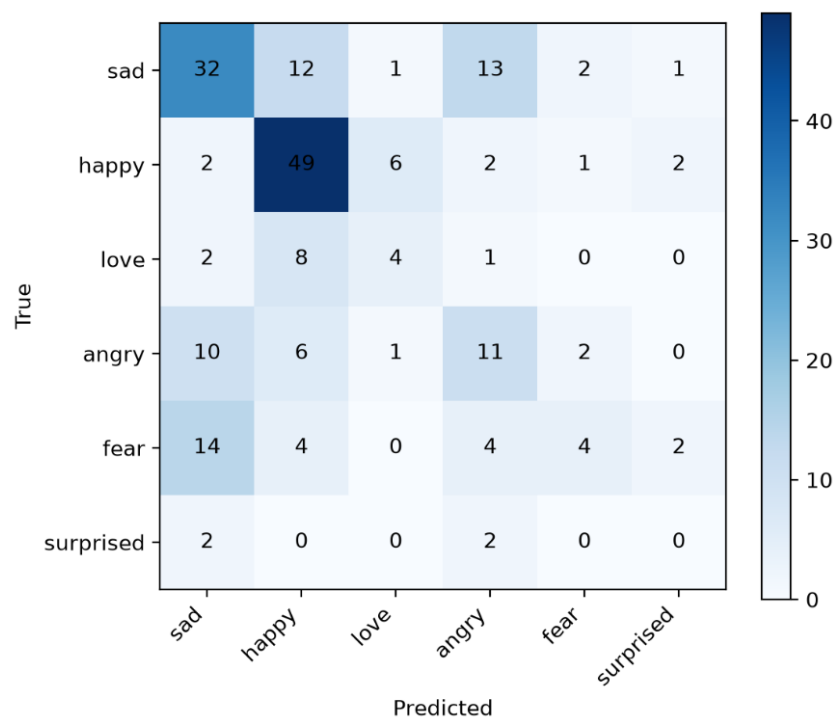
wins on every metric (Acc 0.530 vs. 0.500, macro-F1 0.388 vs. 0.346, ROC-AUC 0.797 vs. 0.790)

Key finding:

"Surprise" (only 4 test examples) scored 0 for both models — absorbed into Sadness/Anger (see confusion matrices, next slide)

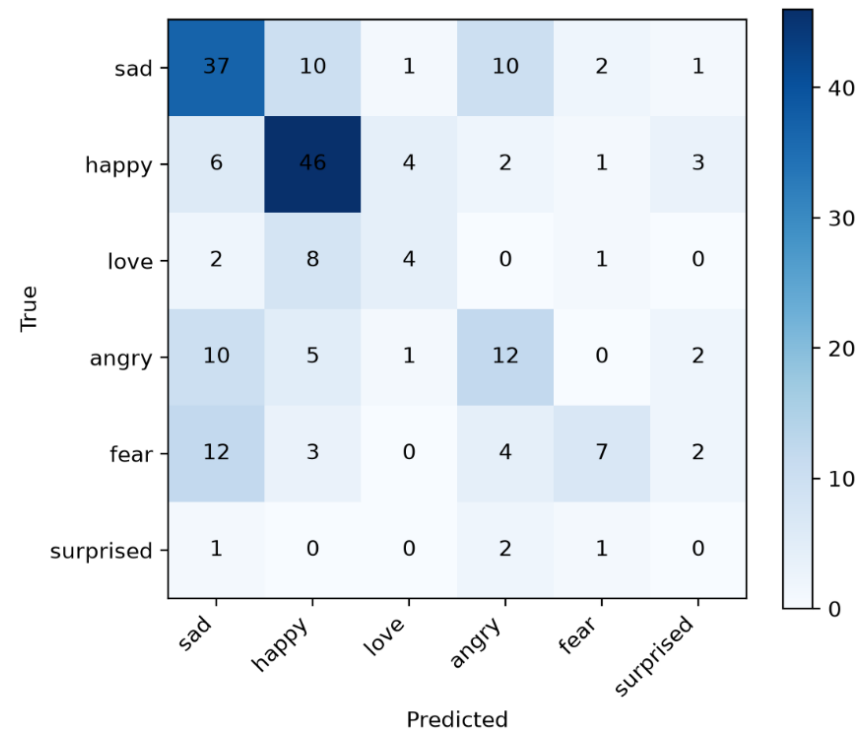
4.1 Confusion Matrices

Confusion Matrix



Logistic Regression

Confusion Matrix



SVM

Fear → Sadness, Love → Joy, and a symmetric Anger ↔ Sadness confusion dominate both models. Surprise (4 examples) is never predicted correctly — a data-volume issue, not a modelling one.

5. DEPLOYMENT & CONCLUSION

From notebook to live web app



- Trained classifier serialised with joblib → emotion_model.pkl
- Streamlit app caches BERT + classifier via @st.cache_resource
- Hosted free on Streamlit Community Cloud, public shareable link



Conclusion

A complete, reproducible pipeline, from raw tweets to an 88.3%-accurate, publicly deployed emotion classifier. Re-run with BERT for a stronger final model.

Corpora Viewer • Embeddings • Model Evaluation • Live Emotion Detection

<https://emotion-detection-ayr8nn66wjtuzydbkezftw.streamlit.app/>

Thank you